

Exploratory Data Analysis: Iris Dataset

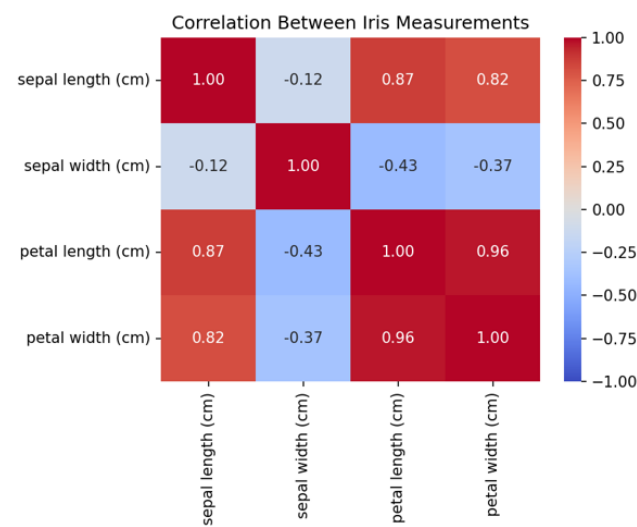
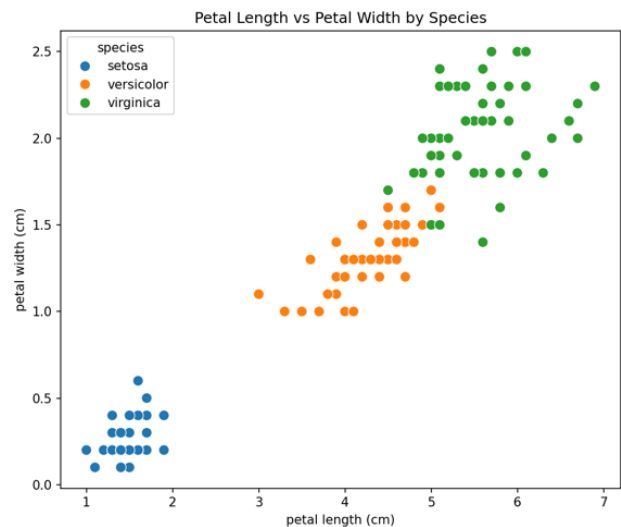
Findings Report | 150 samples, 4 numeric features, 3 species

Data Quality

The dataset is clean: no missing values across any of the 5 columns, and all measurements fall within physically plausible ranges (no negative or zero lengths/widths). One exact duplicate row was found; this is a known characteristic of the classic Iris dataset rather than a data entry error, so it was retained. The three species are perfectly balanced at 50 samples each, meaning no class dominates the visual analysis.

Key Findings

- **Petal measurements separate species far better than sepal measurements.** Petal length and width show almost no overlap between species in box plots, while sepal width overlaps heavily across all three and would be a weak feature on its own.
- **Setosa is the most distinct species.** It forms a completely isolated cluster on petal length/width, separable from the other two species with a single straight line.
- **Versicolor and virginica are more similar to each other** than either is to setosa, with mild overlap in petal measurements around the boundary between the two.
- **Petal length and petal width are highly correlated ($r = 0.96$).** Larger flowers scale proportionally in both dimensions, so the two features carry overlapping information.
- **Sepal width behaves differently from the other three features**, showing a weak negative correlation with all of them — the only feature pair with negative correlation in the dataset.



Left: petal length vs. petal width by species — setosa is fully separated from the other two. Right: correlation matrix — petal length and width are nearly perfectly correlated (0.96).

Conclusion

Petal length and petal width are the most informative features in this dataset for distinguishing iris species and should be prioritized in any downstream classification task. Sepal measurements, particularly sepal width, carry comparatively little species-discriminating signal on their own.